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Photoplethysmography basics

Photoplethysmography (PPG) is a simple and low-cost optical technique that can be used to detect blood volume changes in the microvascular bed of tissue. It is often used non-invasively to make measurements at the skin surface. The PPG waveform comprises a pulsatile (AC) physiological waveform attributed to cardiac synchronous changes in the blood volume with each heart beat, and is superimposed on a slowly varying (DC) baseline with various lower-frequency components attributed to respiration, sympathetic nervous system activity and thermoregulation.

PPG technology is used in a wide range of commercially available medical devices to measure oxygen saturation, blood pressure and cardiac output, assess autonomic function and also detect peripheral vascular disease.

While a pulse oximeter is not able to measure total haemoglobin, TouchHb is designed to do that.

Obstacles

Since non-invasive procedures for the masses is still a field in its infancy, developing the device was not without its share of challenges.

Dr Abhijeet says, "We often joke that we failed 32 times but in reality we probably failed many more times—maybe many times each day! First, the basic science—the core R&D itself—was a challenge. All this 'optical' stuff for a bunch of relatively inexperienced doctors and engineer meant that we were walking in the dark."

Dr Yogesh adds, "There were small nitty gritty details related to production and assembly lines—a millimetre here and a rupee difference there—and suddenly we were struggling both on the technical and costing sides! We had to learn as we went along—how to identify signs of error, rectify and make the hardware reliable. We learnt the best way—by being wrong!"

Roadmap

TouchHb is being produced and sold directly to clinics, which then use it for screening and monitoring of anaemia in their patients. But the team has a bigger plan—to scale up the production from 30-40 a batch to more than 1000 a batch. This will involve putting in place a quality management system, something which will take time.

"Apart from scaling up the production, one of the big steps for us is to partner with different international



TouchHb device

and national health agencies to understand the best way to create an impact with this device. The healthcare ecosystem is a complex thing—protocols are designed for good reasons, and we need to work jointly with experienced public health experts to modify the existing system and make small incremental tweaks in the way point-of-care community health works," says Dr Patil.

The TouchHb Version 1 has already been released. So what next?

Myshkin shares, "As with any technology, you have to keep upgrading, making it better to avoid obsolescence. So watch out for even more feature-rich Version 2, a few months down the road! I would love to tell you about the features we are planning to introduce in subsequent versions, but we ourselves are not sure about which ones will make the cut." ●

The author is a tech correspondent at EY Bengaluru